MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 5: Method of Characteristics: Non-linear Problems

Instructor: Biswarup Biswas

1 Introduction

In Lecture 4, we applied the Method of Characteristics to the linear advection equation. Because the wave speed a was constant, all characteristic lines were parallel, and the solution profile translated across space without any change in shape.

In this lecture, we study the **inviscid Burgers' equation**, where the wave speed depends on the solution u itself. This non-linearity causes characteristics to have different slopes at different points in space. Depending on the initial profile, two distinct physical phenomena arise:

- When characteristics **converge**, the wave profile steepens until the slope becomes vertical, leading to a **shock wave**.
- When characteristics **diverge**, the solution spreads out continuously, forming a **rarefaction wave**.

2 The Inviscid Burgers' Equation

The inviscid Burgers' equation in one dimension is:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0, \qquad x \in \mathbb{R},\ t > 0 \tag{1}$$

with initial condition:

$$u(x, 0) = u_0(x) \tag{2}$$

This is a **quasilinear** first-order PDE. The effective wave speed is $a(u) = u$. Therefore, regions where u is large move faster, while regions where u is small move slower.

3 Applying the Method of Characteristics

3.1 Characteristic Ordinary Differential Equations

Consider a path $x = x(t)$ in the space-time plane. The total rate of change of u along this path is given by the chain rule:

$$\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{dx}{dt}\frac{\partial u}{\partial x} \tag{3}$$

Comparing this expansion with equation (1), we set the path speed equal to the coefficient of $\frac{\partial u}{\partial x}$:

$$\frac{dx}{dt} = u(x(t), t) \tag{4}$$

$$\frac{du}{dt} = 0 \tag{5}$$

3.2 Solving Along the Characteristics

From equation (5), the value of u is **constant** along each characteristic path. If a characteristic line begins at $x = x_0$ at $t = 0$, its value remains equal to the initial value:

$$u(x(t), t) = u(x_0, 0) = u_0(x_0) \tag{6}$$

Since $u = u_0(x_0)$ is constant along the line, equation (4) integrates immediately:

$$\boxed{x(t) = x_0 + u_0(x_0)\, t} \tag{7}$$

Key Property of Burgers' Characteristics

Each characteristic is a straight line in the (x, t) plane starting at x_0. However, unlike the linear advection equation, the slope of each line is determined by the local initial value $u_0(x_0)$. Points with larger u_0 produce lines with larger speed $\frac{dx}{dt}$.

4 Wave Steepening and Breakdown of the Solution

4.1 Evolution of the Spatial Derivative

Because points with larger values of u travel faster than points with smaller values of u, any region where the initial profile is decreasing will steepen over time. To quantify how fast this steepening happens, let us calculate the spatial derivative (slope) $\frac{\partial u}{\partial x}(x, t)$.

Along a characteristic line starting at x_0, the solution is $u(x, t) = u_0(x_0)$ where $x = x_0 + u_0(x_0)t$. Differentiating u with respect to x using the chain rule gives:

$$\frac{\partial u}{\partial x} = u_0'(x_0)\frac{\partial x_0}{\partial x} \tag{8}$$

To find $\frac{\partial x_0}{\partial x}$, we differentiate the position relation $x = x_0 + u_0(x_0)t$ with respect to x:

$$1 = \frac{\partial x_0}{\partial x} + u_0'(x_0)\, t\, \frac{\partial x_0}{\partial x} = \left[1 + u_0'(x_0)\, t\right] \frac{\partial x_0}{\partial x} \tag{9}$$

Solving for $\frac{\partial x_0}{\partial x}$:

$$\frac{\partial x_0}{\partial x} = \frac{1}{1 + u_0'(x_0)\, t} \tag{10}$$

Substituting this back into equation (8), we obtain an exact expression for the slope at any future time:

$$\boxed{\frac{\partial u}{\partial x} = \frac{u_0'(x_0)}{1 + u_0'(x_0)\, t}} \tag{11}$$

4.2 When Does the Slope Become Vertical? (The Breaking Time)

Equation (11) gives us immediate physical insight into the solution behavior:

1. **When $u_0'(x_0) > 0$ (Increasing profile):** The denominator $1 + u_0'(x_0)t > 1$ increases with time. Therefore, the slope magnitude $|\frac{\partial u}{\partial x}|$ decreases, meaning the profile flattens out smoothly.

2. **When $u_0'(x_0) < 0$ (Decreasing profile):** As time increases, the denominator $1 + u_0'(x_0)t$ decreases toward zero. Consequently, the slope $|\frac{\partial u}{\partial x}|$ grows larger and larger: the wave profile **steepens**.

3. **Wave Breaking:** At the exact instant when the denominator reaches zero:

$$1 + u_0'(x_0)\,t = 0 \implies t = -\frac{1}{u_0'(x_0)} \tag{12}$$

the slope becomes vertical: $\frac{\partial u}{\partial x} \to -\infty$.

The earliest time at which a vertical slope appears anywhere in the domain is called the **breaking time** t_s:

$$\boxed{t_s = \min_{x_0,\, u_0'(x_0)<0} \left(-\frac{1}{u_0'(x_0)}\right) = \frac{-1}{\min_{x_0} u_0'(x_0)}} \tag{13}$$

4.3 Worked Example: The Linear Ramp

Consider the linear ramp initial condition:

$$u_0(x) = \begin{cases} 2, & x \le 0 \\ 2 - x, & 0 < x < 1 \\ 1, & x \ge 1 \end{cases} \tag{14}$$

Step 1: Compute the initial slope. On the ramp region $0 < x < 1$, the initial slope is constant: $u_0'(x) = -1$.

Step 2: Calculate the breaking time. Using equation (13):

$$t_s = -\frac{1}{-1} = 1.0 \tag{15}$$

Step 3: Track the profile evolution.

- At $t = 0$: the ramp connects $(x = 0, u = 2)$ to $(x = 1, u = 1)$, with slope -1.
- At $t = 0.5$: the point at $x_0 = 0$ moves to $x = 0 + 2(0.5) = 1.0$. The point at $x_0 = 1$ moves to $x = 1 + 1(0.5) = 1.5$. The slope is now $\frac{1-2}{1.5-1.0} = -2.0$ (steepened).
- At $t = t_s = 1.0$: the point at $x_0 = 0$ moves to $x = 0 + 2(1.0) = 2.0$. The point at $x_0 = 1$ moves to $x = 1 + 1(1.0) = 2.0$. Both endpoints reach $x = 2.0$ simultaneously. The ramp width shrinks to zero, forming a vertical jump at $x_s = 2.0$.

Figure 1 illustrates this wave steepening process at $t = 0$, $t = 0.5$, and $t = 1.0$.

4.4 Characteristic Viewpoint: Focal Point (x_s, t_s)

We can also understand the breaking time geometrically in the space-time (x, t) plane. For the ramp initial condition, the characteristic line from any point $x_0 \in [0, 1]$ is:

$$x(t) = x_0 + (2 - x_0)\,t = x_0(1 - t) + 2t \tag{16}$$

At time $t = 1.0$, the x_0 dependence completely drops out: $x(1.0) = 0 + 2(1.0) = 2.0$, regardless of x_0.

As shown in Figure 2, **all red characteristic lines originating from the ramp focus into the exact same point**:

$$(x_s, t_s) = (2.0, 1.0) \tag{17}$$

This focal point in the (x, t) plane is the geometric representation of the breaking time t_s: the instant where characteristics meet and the smooth solution breaks down.

> **Computational Note**
>
> A Julia implementation for solving Burgers' equation along characteristics and computing the smooth profile up to the breaking time $t \le t_s$ is provided in **Appendix A**.

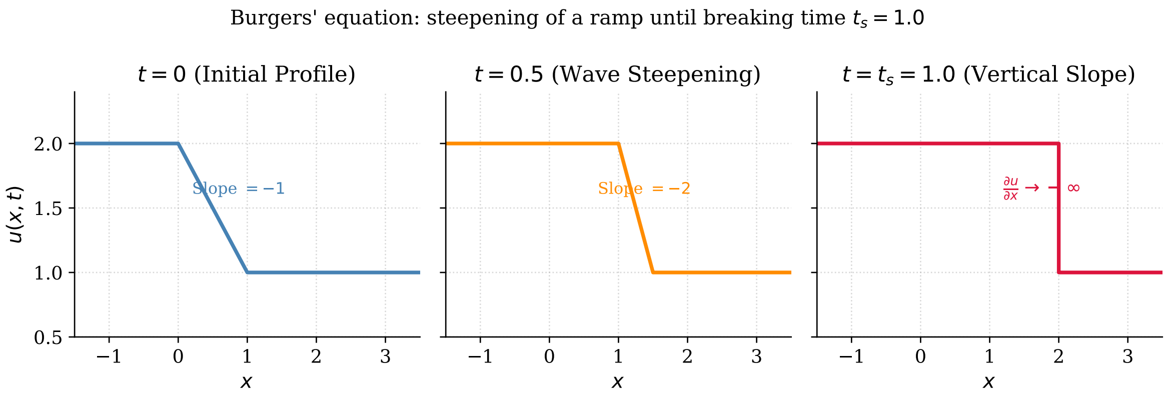

Figure 1: Wave steepening for Burgers' equation with a ramp initial condition. The slope steepens from -1 at $t = 0$, to -2 at $t = 0.5$, and becomes vertical ($\frac{\partial u}{\partial x} \to -\infty$) at the breaking time $t_s = 1.0$.

5 Weak Solutions and the Rankine-Hugoniot Condition

5.1 Conservation Form and Integral Balance

When a vertical slope forms at $t = t_s$, the derivative $\frac{\partial u}{\partial x}$ ceases to exist in the classical sense. To continue the solution past t_s, we rewrite the PDE in **conservation form**:

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) = 0 \implies \frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \tag{18}$$

where $f(u) = \frac{u^2}{2}$ is the flux function.

Integrating this conservation law over any spatial interval $[x_1, x_2]$ and time interval $[t_1, t_2]$ gives the **integral balance**:

$$\boxed{\int_{x_1}^{x_2} u(x, t_2)\, dx - \int_{x_1}^{x_2} u(x, t_1)\, dx = -\int_{t_1}^{t_2} \left[f\big(u(x_2, t)\big) - f\big(u(x_1, t)\big)\right] dt} \tag{19}$$

Definition: Weak Solution

A function $u(x, t)$ that satisfies the integral balance (19) for all control volumes $[x_1, x_2]$ and all time intervals $[t_1, t_2]$ is called a **weak solution**.
Why this works across discontinuities: Notice that equation (19) contains **no derivatives** of $u(x, t)$! The integrals are completely valid even when $u(x, t)$ has sharp jumps (discontinuities).

5.2 The Rankine-Hugoniot Jump Condition

When a discontinuity (shock) moves with speed s, mass conservation across the jump requires the **Rankine-Hugoniot condition**:

$$\boxed{s = \frac{f(u_R) - f(u_L)}{u_R - u_L}} \tag{20}$$

where u_L and u_R denote the states immediately to the left and right of the discontinuity.

For the inviscid Burgers' equation where $f(u) = \frac{u^2}{2}$:

$$s = \frac{\frac{u_R^2}{2} - \frac{u_L^2}{2}}{u_R - u_L} = \frac{(u_R - u_L)(u_R + u_L)}{2(u_R - u_L)} = \frac{u_L + u_R}{2} \tag{21}$$

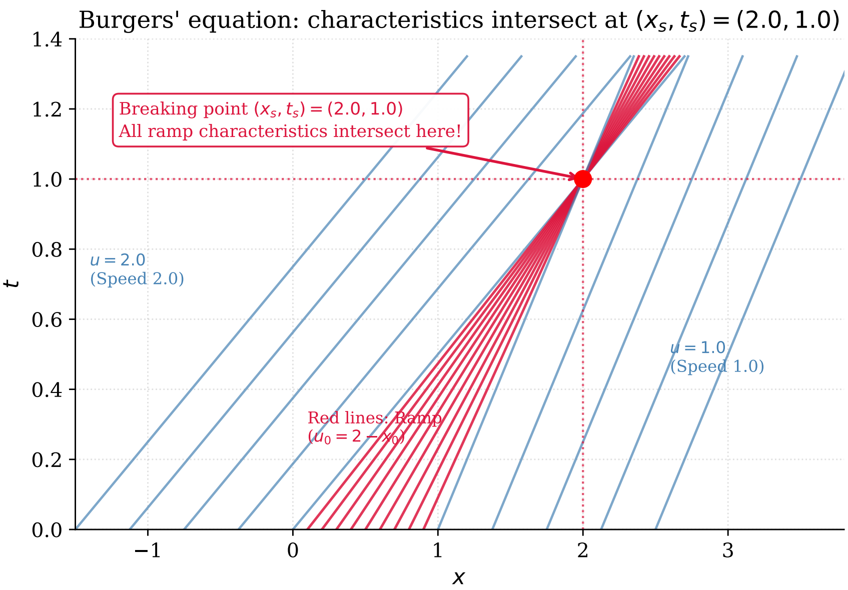

Figure 2: Characteristic lines for the ramp initial condition in Burgers' equation. Faster lines on the left overtake slower lines on the right, causing all red ramp characteristics to focus into the breaking point $(x_s, t_s) = (2.0, 1.0)$.

The shock speed is simply the arithmetic average of the left and right states.

Example: Shock Speed Calculation

For a shock with left state $u_L = 2$ and right state $u_R = 1$, the shock propagation speed is:

$$s = \frac{2+1}{2} = 1.5$$

The discontinuity travels to the right at speed 1.5.

6 Rarefaction Waves

6.1 When Characteristics Diverge

Consider an initial condition where the left state is slower than the right state ($u_L < u_R$):

$$u_0(x) = \begin{cases} u_L, & x < 0 \\ u_R, & x > 0 \end{cases} \qquad (u_L < u_R) \tag{22}$$

Characteristics on the left travel at speed u_L, while characteristics on the right travel at the faster speed u_R. As time advances, these lines **diverge**, creating an expanding gap between $x = u_L t$ and $x = u_R t$.

No characteristics cross, so no shock forms. Instead, a continuous expansion fan, called a **rarefaction wave**, smoothly connects the two states.

6.2 The Rarefaction Fan Solution

Inside the expanding region $u_L t \leq x \leq u_R t$, the characteristic equation $x = ut$ gives $u = x/t$. The complete solution is:

$$\boxed{u(x,t) = \begin{cases} u_L, & x < u_L\, t \\ \dfrac{x}{t}, & u_L\, t \leq x \leq u_R\, t \\ u_R, & x > u_R\, t \end{cases}} \tag{23}$$

Figure 3 shows the diverging characteristics and the smoothly expanding profile.

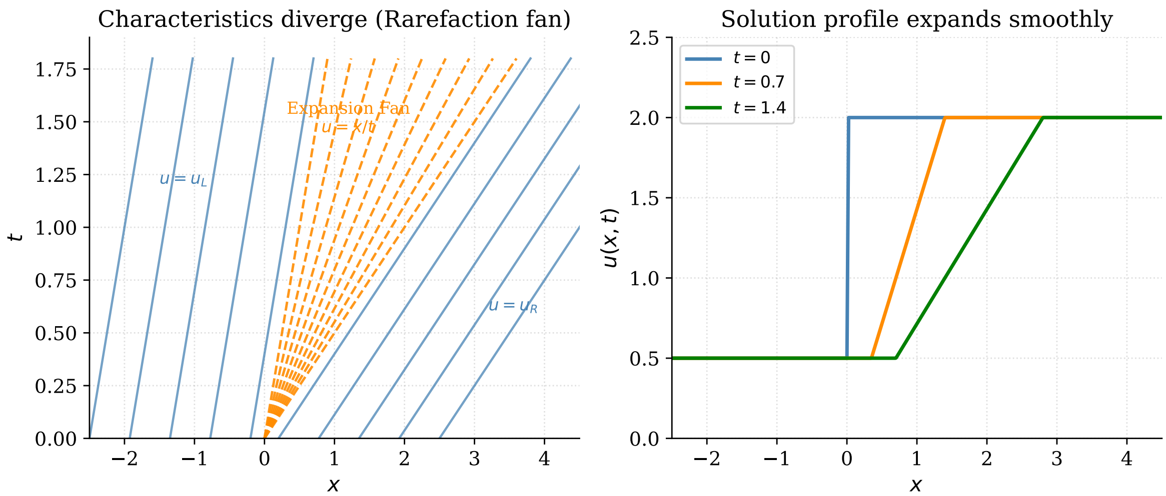

Figure 3: Rarefaction wave ($u_L = 0.5, u_R = 2.0$). Left: characteristic lines fan out from the origin. Right: the solution profile expands smoothly over time, with the fan region growing proportionally to t.

Comparison: Shock vs. Rarefaction

Feature	**Shock Wave**	**Rarefaction Wave**
Initial condition	$u_L > u_R$ (decreasing profile)	$u_L < u_R$ (increasing profile)
Characteristic behavior	Converge and intersect	Diverge and spread apart
Mathematical nature	Discontinuous weak solution	Continuous smooth fan
Propagation speed	$s = \frac{u_L + u_R}{2}$	Fan speed $\frac{x}{t} \in [u_L, u_R]$

7 Exercises

Exercises for Lecture 5

E1. For Burgers' equation with initial condition $u_0(x) = 1 - x^2$ for $|x| \leq 1$ and $u_0(x) = 0$ elsewhere:

(a) Find the region where the initial profile is decreasing.

(b) Determine the minimum initial derivative $\min u_0'(x)$.

(c) Compute the exact breaking time t_s.

E2. Consider the initial condition $u_0(x) = \begin{cases} 1, & x < 0 \\ 3, & x > 0 \end{cases}$:

(a) Explain why this data produces a rarefaction wave rather than a shock.

(b) Write the exact formula for $u(x,t)$ for $t > 0$.

(c) Evaluate the solution at the point $(x = 4, t = 2)$.

E3. A shock wave connects $u_L = 4$ to $u_R = 2$ in Burgers' equation. Find the shock speed s and determine the position of the shock at $t = 3$ if it starts at $x = 0$.

E4. Write the conservation law $\frac{\partial u}{\partial t} + \frac{\partial}{\partial x}(u^3) = 0$ in quasilinear form. What is the characteristic wave speed as a function of u?

Programming Assignment: Method of Characteristics in Julia (Submission Task)

Objective: Implement the Method of Characteristics in Julia to simulate non-linear wave steepening up to the breaking time.

Problem Statement:
Consider the inviscid Burgers' equation $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0$ on the spatial domain $x \in [0, 2\pi]$ with smooth sinusoidal initial data:

$$u(x, 0) = u_0(x) = 1 + \frac{1}{2}\sin(x)$$

Tasks to Complete and Submit:

(a) Analytical Calculation: Compute the initial derivative $u_0'(x)$, find the location of the minimum slope $\min_x u_0'(x)$, and calculate the exact breaking time t_s.

(b) Julia Code Implementation: Using the code in Appendix A as a reference, write a Julia script to solve the characteristic equation $x_0 + u_0(x_0)t = x$ numerically for x_0 using the bisection method on a grid of $N = 200$ points in $[0, 2\pi]$.

(c) Solution Evaluation: Compute and plot (or tabulate) the solution profile $u(x, t)$ at three distinct times:

- $t = 0.0$ (Initial smooth state)
- $t = 1.0$ ($0.5\,t_s$, showing wave steepening)
- $t = 1.95$ ($0.975\,t_s$, just before shock formation)

(d) Observation: Identify the spatial coordinate x around which the wave profile becomes nearly vertical as $t \to t_s$.

Submission Deliverables: Submit your documented Julia script (`.jl` file) along with a brief PDF report showing your analytical derivation and computed results.

A Computational Implementation (Julia)

A.1 Problem Statement

Solve the initial value problem for the inviscid Burgers' equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0, \qquad x \in [-3, 4], \quad t > 0 \tag{24}$$

with the smooth Gaussian initial profile:

$$u(x, 0) = u_0(x) = e^{-x^2} \tag{25}$$

A.2 Analytical Preparation

1. **Initial derivative:** $u_0'(x) = -2xe^{-x^2}$.
2. **Steepest downward slope:** Setting $\frac{d^2u_0}{dx^2} = (-2 + 4x^2)e^{-x^2} = 0$ gives $x^* = \frac{1}{\sqrt{2}} \approx 0.7071$.
3. **Minimum derivative value:** $u_0'(x^*) = -\sqrt{2}e^{-1/2} = -\sqrt{\frac{2}{e}} \approx -0.8578$.

4. **Analytical breaking time:**

$$t_s = \frac{-1}{\min u_0'(x)} = \sqrt{\frac{e}{2}} \approx 1.1658 \tag{26}$$

A.3 Julia Code

The following script solves the implicit characteristic equation $x = x_0 + u_0(x_0)t$ for x_0 using the bisection method, and evaluates $u(x,t) = u_0(x_0)$ at any time $t < t_s$:

Listing 1: Julia Solver: Method of Characteristics for Burgers' Equation ($t \leq t_s$)

```julia
# ===================================================================
# Solving Burgers' Equation with u0(x) = exp(-x^2) up to t <= t_s
# ===================================================================

# 1. Define initial condition and its derivative
u0(x) = exp(-x^2)
u0_prime(x) = -2.0 * x * exp(-x^2)

# 2. Compute the exact breaking time: t_s = -1 / min(u0'(x))
function compute_breaking_time()
    x_test = range(-3.0, 3.0, length=1000)
    min_slope = minimum([u0_prime(xi) for xi in x_test])
    t_s = -1.0 / min_slope
    return t_s
end

# 3. For given (x, t), find initial coordinate x0: x0 + u0(x0)*t = x
function find_x0(x, t; tol=1e-7, max_iter=100)
    a = x - 3.0 * (t + 1.0)
    b = x + 3.0 * (t + 1.0)
    g(x0) = x0 + u0(x0) * t - x

    for _ in 1:max_iter
        c = (a + b) / 2.0
        if abs(g(c)) < tol || (b - a) / 2.0 < tol
            return c
        end
        if g(a) * g(c) < 0.0
            b = c
        else
            a = c
        end
    end
    return (a + b) / 2.0
end

# 4. Evaluate solution u(x,t) = u0(x0) on a spatial grid
function solve_burgers(x_grid, t)
    u_sol = zeros(length(x_grid))
    for (i, xi) in enumerate(x_grid)
        x0 = find_x0(xi, t)
        u_sol[i] = u0(x0)
    end
    return u_sol
end

# Execute calculation:
t_s = compute_breaking_time()
println("Analytical Breaking Time t_s = ", round(t_s, digits=4))

# Evaluate profile at t = 0, t = 0.5*t_s, and t = 0.99*t_s
```

```julia
for t_val in [0.0, 0.5 * t_s, 0.99 * t_s]
    u_vals = solve_burgers([0.0, 1.0, 2.0], t_val)
    println("t = $(round(t_val, digits=3)): u(0.0) = $(round(u_vals[1], digits
        =4)), u(1.0) = $(round(u_vals[2], digits=4)), u(2.0) = $(round(u_vals
        [3], digits=4))")
end
```

A.4 Computed Output

Julia Terminal Output

```
Analytical Breaking Time t_s = 1.1658
t = 0.000:  u(0.0) = 1.0000, u(1.0) = 0.3679, u(2.0) = 0.0183
t = 0.583:  u(0.0) = 0.8032, u(1.0) = 0.7085, u(2.0) = 0.0191
t = 1.154:  u(0.0) = 0.6096, u(1.0) = 0.9823, u(2.0) = 0.0201
```

References

[1] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems.* Cambridge University Press, Cambridge, UK, 2002.

[2] Eleuterio F. Toro. *Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction.* Springer-Verlag, Berlin, Heidelberg, 3rd edition, 2009.